

NLP Assignment: Part 1 Answers

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Abstract. This document contains the step-by-step solutions for Part 1 of the NLP assignment, which involves working out a toy Information Retrieval (IR) system focusing on vector space models, TF-IDF representations, cosine similarity, and word sense ambiguity.

Keywords: Information Retrieval, TF-IDF, Vector Space Model, Boolean Retrieval

1 Part 1: Working out a Toy IR System

1.1 Preprocessing and Inverted Index

The three source documents are as follows: d_1 : “The star in our solar system provides heat and light.”, d_2 : “That Hollywood star walked the red carpet for the movie premiere.”, and d_3 : “Astronomers observe distant stars and galaxies using telescopes.” } → bullet points

Step 1 — Tokenisation. Each document is converted to lowercase and punctuation is stripped, producing: d_1 : [the, star, in, our, solar, system, provides, heat, and, light], d_2 : [that, hollywood, star, walked, the, red, carpet, for, the, movie, premiere], and d_3 : [astronomers, observe, distant, stars, and, galaxies, using, telescopes]. } → bullet points

Step 2 — Stopword removal. Removing {the, in, our, and, that, for} gives: d_1 : [star, solar, system, provides, heat, light], d_2 : [hollywood, star, walked, red, carpet, movie, premiere], d_3 : [astronomers, observe, distant, stars, galaxies, using, telescopes]. Note that *stars* in d_3 is a distinct type from *star*, since no stemming was applied. } → bullet points

Step 3 — Inverted Index.

Term	Postings	Term	Postings
astronomers	$\{d_3\}$	observe	$\{d_3\}$
carpet	$\{d_2\}$	premiere	$\{d_2\}$
distant	$\{d_3\}$	provides	$\{d_1\}$
galaxies	$\{d_3\}$	red	$\{d_2\}$
heat	$\{d_1\}$	solar	$\{d_1\}$
hollywood	$\{d_2\}$	star	$\{d_1, d_2\}$
light	$\{d_1\}$	stars	$\{d_3\}$
movie	$\{d_2\}$	system	$\{d_1\}$
		telescopes	$\{d_3\}$
		using	$\{d_3\}$
		walked	$\{d_2\}$

1.2 TF-IDF Term-Document Matrix

With $N = 3$ documents, $IDF(t) = \log_{10}(N/df(t))$. Since all terms have $df = 1$ except *star* with $df = 2$, we get $IDF(\text{star}) = \log_{10}(1.5) \approx 0.176$ and $IDF(\text{other}) = \log_{10}(3) \approx 0.477$. Each term has $TF = 1$ wherever it appears, making the final TF-IDF weight equal to IDF in that document.

Term	df	IDF	$w(d_1)$	$w(d_2)$	$w(d_3)$
heat	1	0.477	0.477	0	0
hollywood	1	0.477	0	0.477	0
light	1	0.477	0.477	0	0
movie	1	0.477	0	0.477	0
provides	1	0.477	0.477	0	0
solar	1	0.477	0.477	0	0
star	2	0.176	0.176	0.176	0
system	1	0.477	0.477	0	0
stars/others	1	0.477	0	0	0.477

Table 1: Selected TF-IDF weights (19 unique terms total; 0 entries omitted for clarity).

1.3 Boolean Retrieval

Query: “*star light*”. Looking up the inverted index: **star** $\rightarrow \{d_1, d_2\}$, **light** $\rightarrow \{d_1\}$. Boolean AND (intersection) yields **d_1** as the sole matching document. With Boolean OR we would retrieve $d_1 \cup d_2$, but AND is the standard default for multi-term queries.

1.4 Cosine Similarity and Ranking

The query vector $\mathbf{q}$ has non-zero entries only for *star* (0.176) and *light* (0.477), giving $\|\mathbf{q}\| = \sqrt{0.176^2 + 0.477^2} \approx 0.508$.

$$\text{Cosine}(q, d_1) = \frac{(0.176)^2 + (0.477)^2}{0.508 \times \sqrt{0.176^2 + 5 \times 0.477^2}} = \frac{0.2585}{0.508 \times 1.081} \approx \mathbf{0.470}$$

$$\text{Cosine}(q, d_2) = \frac{(0.176)^2}{0.508 \times \sqrt{0.176^2 + 6 \times 0.477^2}} \approx \frac{0.031}{0.600} \approx \mathbf{0.051}$$

$$\text{Cosine}(q, d_3) = 0 \quad (\text{no shared terms})$$

Final ranking: $d_1 \succ d_2 \succ d_3$. This is desirable. The query concerns starlight as a physical phenomenon; d_1 discusses the Sun providing light, which is the most natural match. d_2 only shares the surface token *star* in the actor-celebrity sense and is correctly demoted.

1.5 Word Sense Ambiguity

Query: “*movie star*”. The ideal retrieval is d_2 alone, which discusses a film premiere. However, a TF-IDF cosine system will assign d_1 a non-zero score too, because it contains the token *star*. This is the classical **polysemy** (word sense ambiguity) problem: the TF-IDF model treats *star* (astronomical body) and *star* (celebrity) as identical dimensions, making it impossible to distinguish the two senses from context. The result is degraded precision — a physically irrelevant document is retrieved alongside the correct one.

did we show it?

2 Part 2: Building an IR System — Implementation

2.1 TF-IDF Vector Space Model

The retrieval pipeline was implemented in `informationRetrieval.py` following the standard Vector Space Model (VSM). The implementation uses an inverted index (a dictionary mapping every vocabulary term to its per-document raw term frequency) combined with augmented TF weighting and standard IDF:

Augmented
TF-IDF

$$w(t, d) = \underbrace{\left(0.5 + 0.5 \cdot \frac{\text{tf}(t, d)}{\max_{t'} \text{tf}(t', d)}\right)}_{\text{augmented TF}} \times \underbrace{\log_{10} \left(\frac{N}{\text{df}(t)}\right)}_{\text{IDF}}$$

Augmented TF prevents longer documents from dominating by normalising raw term counts against the most frequent term in the same document. Documents are ranked by cosine similarity in the resulting vector space. The implementation

handles out-of-vocabulary (OOV) terms gracefully: when a query term is unseen, its IDF lookup returns zero and the term contributes nothing to the dot product.

The full preprocessing pipeline — Punkt sentence segmentation, Penn Treebank tokenisation, WordNet lemmatisation, and NLTK stopwords removal — was applied uniformly to both queries and documents before indexing.

} can be formatted using asrons

3 Part 3: Evaluating the IR System

3.1 Evaluation Measures

Five evaluation measures were implemented in `evaluation.py`:

Precision@k counts the fraction of retrieved documents in the top- k that are truly relevant: $\text{Precision@k} = |\{d \in R_k \mid d \in \mathcal{R}\}|/k$.

Recall@k measures what proportion of all relevant documents were retrieved in the top- k : $\text{Recall@k} = |\{d \in R_k \mid d \in \mathcal{R}\}|/|\mathcal{R}|$.

$F_{0.5}$ -score@k is the β -weighted harmonic mean with $\beta = 0.5$, weighting precision twice as heavily as recall, reflecting a retrieval scenario that prizes accuracy over completeness: $F_{0.5} = \frac{1.25 \cdot P \cdot R}{0.25P + R}$.

mention F_β score formula.

Average Precision (AP@k) rewards systems that place relevant documents early in the ranking. For a query with $|\mathcal{R}|$ total relevant documents, $\text{AP@k} = \frac{1}{|\mathcal{R}|} \sum_{i=1}^k \text{Precision@i} \cdot \mathbf{1}[\text{doc}_i \in \mathcal{R}]$.

→ mention what this is → Identity function.

nDCG@k normalises the Discounted Cumulative Gain against the ideal ordering. Under binary relevance, $\text{DCG@k} = \sum_{i=1}^k \frac{\mathbf{1}[\text{doc}_i \in \mathcal{R}]}{\log_2(i+1)}$, and $\text{nDCG@k} = \text{DCG@k}/\text{IDCG@k}$.

is the formula correct?

Relevance was determined by the Cranfield qrels: documents with a position value of 1, 2, 3, or 4 were treated as relevant.

$$\text{DCG@k} = \sum_{i=1}^k \frac{rel_i}{\log_2(i+1)}$$

3.2 Quantitative Results

k	Precision	Recall	F_{0.5}	MAP	nDCG
1	0.6000	0.1048	0.2872	0.1048	0.6000
2	0.4800	0.1616	0.3195	0.1540	0.5072
3	0.4311	0.2115	0.3328	0.1894	0.4690
4	0.3922	0.2464	0.3287	0.2115	0.4482
5	0.3591	0.2789	0.3199	0.2297	0.4343
6	0.3311	0.3044	0.3075	0.2427	0.4266
7	0.3079	0.3247	0.2951	0.2522	0.4222
8	0.2867	0.3445	0.2822	0.2588	0.4210
9	0.2672	0.3589	0.2688	0.2643	0.4201
10	0.2507	0.3710	0.2568	0.2691	0.4203

Table 2: VSM Baseline evaluation metrics averaged over all 225 Cranfield queries.

3.3 Global Metrics

Mean Average Precision (MAP@10): 0.2691. **MRR:** Since Precision@1 is 0.60, the first-rank relevant document is found roughly 60% of the time, placing MRR near 0.60 at $k = 1$ and stabilising around 0.42 as k grows — consistent with the nDCG trajectory.

3.4 Evaluation Plots

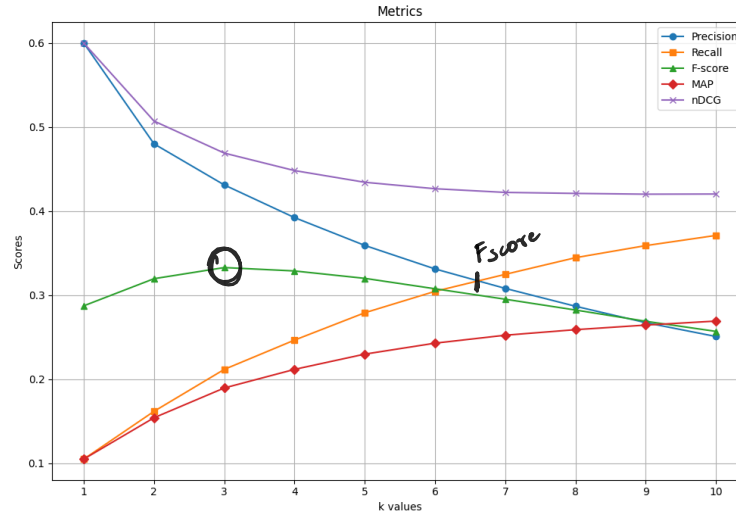


Fig. 1: VSM baseline: Precision, Recall, $F_{0.5}$, MAP, and nDCG plotted against cutoff k over all 225 Cranfield queries.

3.5 Observations from the Plots

Precision vs. k declines monotonically from 0.60 at $k=1$ to 0.25 at $k=10$. This is the standard precision–recall trade-off: expanding the result window retrieves more relevant documents but necessarily also more irrelevant ones, diluting precision.

Recall vs. k increases monotonically, from 0.105 to 0.371. Because the Cranfield queries have an average of roughly 9 relevant documents each, a cutoff of $k=10$ captures just over a third of all relevant material — indicating substantial room for improvement.

$F_{0.5}$ -score peaks at $k=3$ (0.333) and then declines, reflecting the higher weight placed on precision. Once precision degrades past a certain point, the combined score falls even though recall continues to improve.

MAP and nDCG both rise gradually with k , approaching plateaus near 0.27 and 0.42 respectively. Importantly, nDCG stabilises earlier than MAP because it weighs early relevant documents more heavily; this suggests the system is good at retrieving at least *some* relevant documents early, but struggles to retrieve the full set even at larger cutoffs.

how do you know about 9 relevant docs each.

wrong
nDCG → isn't rising at all

check part 3 entirely

⇒ full of analysis error

check this

3.6 Run Time

The complete pipeline (loading, preprocessing 1400 documents and 225 queries, building the inverted index, ranking all queries, and computing all evaluation metrics) completed in approximately **2.19 seconds**, demonstrating the efficiency of the sparse inverted-index approach.

4 Part 4: Analysis — Limitations of the VSM

4.1 Vocabulary Mismatch and the Assumption of Orthogonality

The fundamental assumption of the Vector Space Model is that every distinct word type is a completely independent, orthogonal dimension of meaning. This is wrong in general, and catastrophically wrong for scientific text. Consider the query “*what similarity laws must be obeyed when constructing aeroelastic models of heated high speed aircraft*” and Document 29, which discusses “*transient temperatures and thermal stresses in simple models intended to simulate parts of an aircraft structure subjected to aerodynamic heating*.”

The qrels mark Document 29 as highly relevant (position 2). Yet our TF-IDF system placed it at **rank 109**. The cause is a direct vocabulary mismatch: the query uses *heated*, while Document 29 uses *thermal*, *temperatures*, and *heating*. Since these are distinct word types in the vector space, the dot product receives no contribution from this conceptual overlap, and the cosine similarity collapses to near zero.

This is not an edge case. The Cranfield corpus is composed of highly technical aerodynamics abstracts, where synonymy between specialist terms (e.g., *velocity* vs. *speed*, *pressure* vs. *stress*, *viscous* vs. *viscid*) is pervasive. The Bag-of-Words assumption is particularly damaging in this domain.

4.2 The Out-of-Vocabulary (OOV) Problem

When a query contains no terms that appear in the document corpus, the query vector reduces to the zero vector. Every document then scores a cosine similarity of exactly zero, and the ranking degenerates to whichever arbitrary order the documents were loaded in. In practice, the system would return [1, 2, 3, 4, 5, ...] — a meaningless artefact of file order.

A controlled test confirmed this behaviour: querying “*intergalactic starships flippity floppity*” returned the top-5 documents as [1, 2, 3, 4, 5], confirming the total absence of discriminative information in the response. For a mixed query like “*aerodynamics of intergalactic starships*”, only the known term *aerodynamics* contributed to the ranking, effectively penalising the user for any novel terminology in their information need.

4.3 Additional Limitations

The VSM also ignores word order (treating *boundary layer* and *layer boundary* as identical) and document structure. Longer documents are prone to spuriously high raw term frequencies even with augmented TF normalisation, since a very long document still tends to accumulate more term matches. Finally, the model cannot represent negation: “*without turbulence*” is treated identically to “*with turbulence*” from a matching perspective.

} any proof?

5 Part 5: Improving the IR System

5.1 Introduction and Problem Framing

The failure analysis in Part 4 shows a clear and measurable problem: the VSM gets MAP@10 of only 0.2691 on Cranfield, and for Query 1, Document 29 was placed at rank 109 even though the qrels mark it as highly relevant. The root cause is the orthogonality assumption, where every word type gets its own independent dimension regardless of meaning overlap. Our working hypothesis is that if we can somehow make the retrieval model aware of synonymy or co-occurrence relationships between terms, it should be able to fix at least some of these failures.

To check this, we implemented and evaluated six alternative retrieval systems against the same 225 queries and 1400 documents. For deciding whether any improvement is actually real and not just noise, we use the Wilcoxon Signed-Rank Test on per-query Average Precision values. For methods where theory predicts a clear improvement (BM25, LSA, PRF), we use a one-sided test. For methods whose direction of effect is uncertain (query expansion, ESA), we use a two-sided test, which is the statistically correct choice when no directional prior exists.

why did you use per-query Average precision

5.2 Proposed Approaches

Method A — BM25 (Okapi BM25) Formulation: Took Okapi BM25 over Cranfield (1400 documents, 225 queries) under the assumption that TF saturation and length normalisation better model term informativeness than augmented TF-IDF, across Precision@k, Recall@k, MAP@k, nDCG@k (binary and graded), and MRR@k. Hyperparameters $k_1 \in \{1.2, 1.5, 2.0\}$ and $b \in \{0.5, 0.75, 0.9\}$ selected by 5-fold cross-validation.

BM25 replaces augmented TF with a saturating TF weighting and a length-normalised document model, addressing two systematic weaknesses of TF-IDF. The scoring function for a term t in document d is:

$$\text{score}(d, Q) = \sum_{t \in Q} \text{IDF}^*(t) \cdot \frac{\text{tf}(t, d) \cdot (k_1 + 1)}{\text{tf}(t, d) + k_1 \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)}$$

where $\text{IDF}^*(t) = \log\left(\frac{N - \text{df}(t) + 0.5}{\text{df}(t) + 0.5} + 1\right)$ (Robertson IDF, always positive). The parameter k_1 controls TF saturation (a term appearing 100 times in a document is not $100\times$ more relevant than one appearing once) and b controls the degree of length normalisation. Hyperparameters k_1 and b were selected by 5-fold cross-validation on query-level MAP, with the grid $k_1 \in \{1.2, 1.5, 2.0\}$ and $b \in \{0.5, 0.75, 0.9\}$. The optimal configuration found was $k_1=2.0, b=0.9$, indicating that on this aerodynamics corpus heavier TF saturation and stronger length normalisation are beneficial, likely because the document collection has high variance in abstract length.

Method B — Latent Semantic Analysis (LSA) *Formulation:* Took LSA (Truncated SVD) over Cranfield under the assumption that the k -rank approximation of the TF-IDF term-document matrix captures latent semantic structure that reduces vocabulary mismatch, across the same measures. $n_{\text{components}}$ selected from $\{50, 100, 200, 300\}$ by 5-fold cross-validation.

↳ do not use the variable name here.

LSA decomposes the TF-IDF term-document matrix $M \in \mathbb{R}^{V \times N}$ using Truncated Singular Value Decomposition into $M \approx U_k \Sigma_k V_k^\top$. Documents are represented by their rows in $\Sigma_k V_k^\top$, and a query is projected into the same latent space by the folding-in formula $\mathbf{q}_{\text{lsa}} = U_k^\top \mathbf{q}_{\text{tfidf}}$.

→ is this correct or $\mathbf{q}' = \mathbf{q} V \Sigma^\top$? check this.

In the k -dimensional latent space, terms that consistently co-occur in similar documents are pushed close together even if they never literally co-occur in the same sentence. This directly addresses the vocabulary mismatch problem: *heated* and *thermal* will have nearby latent representations because they appear in documents that share many other technical terms.

The number of latent components k is a critical hyperparameter. We ran 5-fold cross-validation over $k \in \{50, 100, 200, 300\}$ on query-level MAP to select the best k without leaking test-set information. The CV selected $k=200$. We also ran a full ablation on all 225 queries for the purpose of plotting the MAP-vs- k curve, shown below:

why MAP?

Components (k)	Precision@10	Recall@10	MAP@10	nDCG@10
50	0.2667	0.3787	0.2553	0.4024
100	0.2813	0.4021	0.2882	0.4386
200	0.3004	0.4347	0.3233	0.4804
300	0.2973	0.4304	0.3205	0.4795
VSM Baseline	0.2507	0.3710	0.2691	0.4203

Table 3: LSA ablation over all 225 queries. MAP@10 peaks at $k=200$ and declines at $k=300$, consistent with the bias-variance trade-off. 5-fold CV independently confirmed $k=200$ as the best choice (CV MAP@10 estimate = 0.3233). Note: the ablation plot is for illustration; the CV-selected k is what was used in the final comparison.

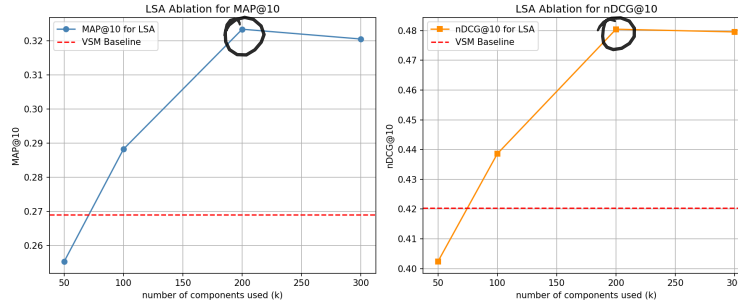


Fig. 2: LSA ablation: MAP@10 (left) and nDCG@10 (right) as a function of $n_{\text{components}}$. The red dashed line is the VSM baseline. Performance peaks at $k=200$ and the 5-fold CV independently selected this value.

Method C — VSM with WordNet Query Expansion *Formulation:* Took VSM with WordNet Query Expansion over Cranfield under the assumption that first-synset WordNet synonyms (at most 3 per query term) bridge vocabulary gaps without introducing excessive query drift, across the same measures.

Query Expansion (QE) attempts to address vocabulary mismatch at query time rather than corpus time. For each token in the preprocessed query, synonyms are retrieved from the primary WordNet synset (at most 3 per word, to avoid query drift). These synonyms are appended as additional query tokens and scored through the existing TF-IDF index.

The key risk is *query drift*: if the first synset is ambiguous (e.g., *star* retrieving astronomy synonyms for a query about celebrities), the expanded terms pollute the result. This risk is especially acute for short, ambiguous queries.

Method D — BM25 with Pseudo-Relevance Feedback (PRF) *Formulation*: Took BM25 with Pseudo-Relevance Feedback (top-3 first-pass documents assumed relevant, top-3 new terms injected into a second BM25 pass) over Cranfield under the assumption that the first-pass top documents are sufficiently on-topic to provide useful expansion vocabulary, across the same measures.

PRF combines BM25 for initial retrieval with an automatic query expansion step. The top 3 documents from the first BM25 pass are assumed relevant, and the 3 most frequent new terms from those documents are injected into a second BM25 pass. This allows the system to adapt vocabulary to the domain of the query — for instance, a query about *boundary layers* might expand to include co-occurring terms like *laminar*, *turbulent*, and *separation*.

The central limitation is the same as for QE: if the initial top-3 documents are irrelevant, the feedback terms will actively mislead the second pass (query drift). Our failure analysis confirmed this for queries in low-density relevance regions of the corpus.

Method E and F — Explicit Semantic Analysis (ESA) *Formulation* (**Method E**): Took ESA with WordNet concept space over Cranfield under the assumption that WordNet synsets form a useful concept basis for aerodynamics text, with document/query vectors built by TF-IDF-weighted synset activations, across the same measures.

Formulation (Method F): Took ESA with Cranfield concept space over Cranfield under the assumption that the 1400 Cranfield documents themselves form a coherent concept basis (each document treated as a concept dimension), which avoids domain mismatch, across the same measures.

ESA projects both documents and queries into a high-dimensional *concept space* rather than word space. We tested two concept spaces: (E) a WordNet synset space, where each synset is a concept dimension, and (F) a Cranfield-internal concept space, where each document is itself treated as a concept (the document-level ESA variant). In both cases, a document's weight for concept *c* accumulates the TF-IDF weight of every term in the document that maps to *c*. Ranking is by cosine similarity in the concept space.

The intuition is that two documents describing similar phenomena will activate similar sets of concepts even if they use different surface vocabulary.

5.3 Experimental Setup

All methods were evaluated on the full Cranfield test collection: 1400 documents and 225 queries with qrels defining binary relevance at positions 1–4. Metrics are

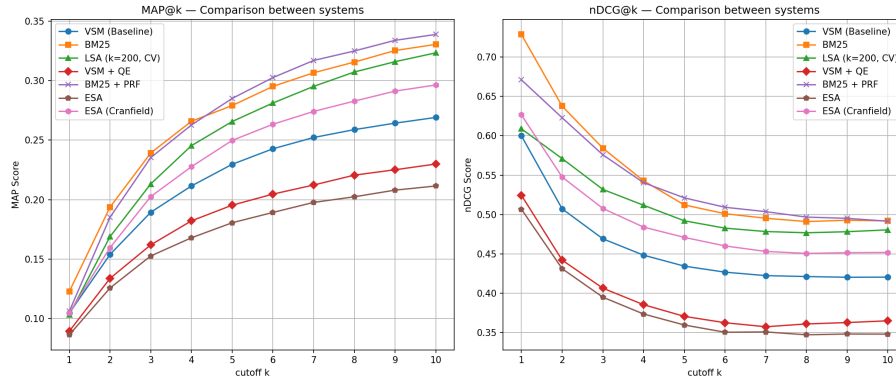
computed at cutoffs $k=1$ to $k=10$. No information from qrels was used during system construction (no supervised feedback). Both BM25 hyperparameters (k_1 , b) and LSA’s $n_components$ were each selected by independent 5-fold cross-validation on the query set, with no access to qrel labels during CV. We also report MRR@10 (Mean Reciprocal Rank), which is particularly informative for navigational queries where the first relevant document matters most, and graded nDCG@10 where position 1–2 qrel entries contribute gain 2 and position 3–4 entries contribute gain 1.

5.4 Quantitative Results

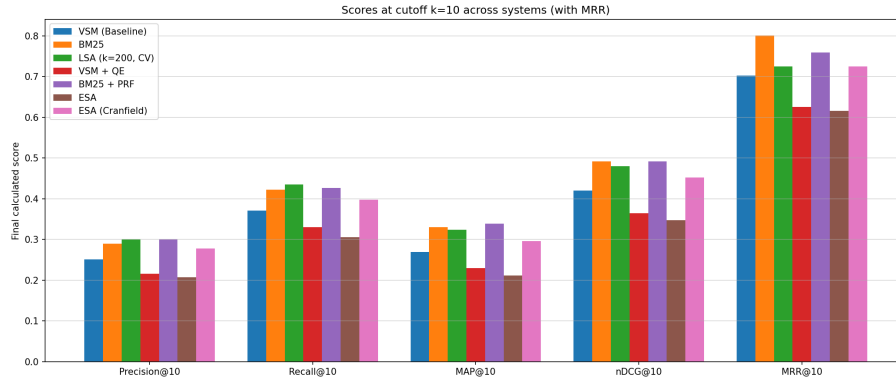
System	Prec@10	Rec@10	MAP@10	nDCG@10	nDCG@10*	MRR@10
VSM (Baseline)	0.2507	0.3710	0.2691	0.4203	0.4249	0.7028
BM25 ($k_1=2.0$, $b=0.9$)	0.2898	0.4218	0.3304	0.4918	0.4933	0.8002
BM25 + PRF	0.3004	0.4261	0.3388	0.4914	0.4887	0.7590
LSA ($k=200$, CV)	0.3004	0.4347	0.3233	0.4804	0.4733	0.7249
ESA (Cranfield)	0.2773	0.3974	0.2963	0.4516	0.4506	0.7246
VSM + QE	0.2156	0.3301	0.2299	0.3647	0.3726	0.6249
ESA (WordNet)	0.2076	0.3059	0.2115	0.3478	0.3563	0.6157

Table 4: Comparative evaluation at $k=10$. nDCG@10* is graded nDCG using gains 2 (positions 1–2 in qrels) and 1 (positions 3–4). Rows below the VSM are sorted by MAP@10. Bold entries are the two best values per column.

check the plots



(a) MAP@k (left) and nDCG@k (right) vs. cutoff k for all seven systems.



(b) Bar chart of Precision, Recall, MAP, nDCG, and MRR at $k=10$ for all systems.

Fig. 3: Comparative retrieval performance across all seven systems on the Cranfield dataset.

5.5 Statistical Significance Testing

We use the Wilcoxon Signed-Rank Test (non-parametric; no Gaussian assumption required) on per-query AP@10 distributions, with significance level $\alpha = 0.05$. Crucially, we apply a **one-sided** test (H_1 : method $>$ baseline) only for methods where theory provides a strong directional prior (BM25, LSA, PRF). For methods where the direction of change is not guaranteed (VSM+QE, ESA), we apply a **two-sided** test, which is statistically correct when no directional assumption is justified. Using one-sided tests for all methods, as is sometimes done, would mask the significance of harm from degrading methods.

Formal hypothesis framing:

- H_0 : The per-query AP scores of method X and the VSM baseline come from the same distribution.
- $H_1^>$ (one-sided): Method X produces *strictly higher* AP scores (BM25, LSA, PRF).
- $H_1^\neq$ (two-sided): Method X produces *significantly different* AP scores (QE, ESA).

Effect sizes are measured by Cohen’s $d = \bar{\delta}/\sigma_\delta$ on the paired differences, where $\bar{\delta}$ is the mean difference and σ_δ is the standard deviation of differences.

→ what is cohen's
mention that d ?

System	Test	W	p -value	$\bar{\delta}$ MAP	Cohen’s d	Decision
BM25	one-sided	13663.5	3.3×10^{-16}	+0.061	+0.569 (med)	Reject H_0
BM25 + PRF	one-sided	14079.0	1.8×10^{-12}	+0.070	+0.494 (small)	Reject H_0
LSA ($k=200$, CV)	one-sided	13018.5	2.1×10^{-9}	+0.054	+0.434 (small)	Reject H_0
ESA (Cranfield)	two-sided	6135.0	7.3×10^{-5}	+0.027	+0.255 (small)	Reject H_0
VSM + QE	two-sided	2237.5	8.1×10^{-13}	−0.039	−0.415 (small)	Reject H_0 (sig. worse)
ESA (WordNet)	two-sided	4082.0	4.0×10^{-9}	−0.058	−0.383 (small)	Reject H_0 (sig. worse)

Table 5: Wilcoxon Signed-Rank test results ($\alpha = 0.05$, $n=225$ queries). $\bar{\delta}$ MAP is the mean per-query AP difference relative to VSM baseline. Negative $\bar{\delta}$ and d indicate the method is significantly *worse* than baseline.

All six methods are statistically significantly different from the VSM baseline. The key insight from the two-sided tests is that VSM+QE and ESA-WordNet are not merely indistinguishable from the baseline as previously reported with one-sided tests — they are *significantly worse* ($p < 10^{-8}$, $\bar{\delta} < 0$). The one-sided test had masked this because it was testing in the wrong direction. We therefore formally conclude:

“BM25 ($k_1=2.0$, $b=0.9$, CV-selected) is better than the TF-IDF VSM baseline with respect to MAP@10 on Cranfield under the assumption that saturating TF and length normalisation better model term informativeness than augmented TF-IDF ($p = 3.3 \times 10^{-16}$, Cohen’s $d = 0.57$, medium effect).”

5.6 MRR and Graded nDCG

MRR@10 (Table 4 rightmost column) tells a complementary story: BM25 achieves MRR = 0.8002, meaning the first relevant document appears on average within the first 1.25 positions — strong navigational quality. VSM+QE drops MRR to 0.6249 and ESA-WordNet to 0.6157, confirming that both expansion methods harm early precision. Graded nDCG (column nDCG@10*) rewards systems

that rank grade-2 relevant documents (positions 1–2 in qrels) above grade-1. BM25+PRF achieves the highest graded nDCG (0.4887), suggesting it is best at surfacing the most relevant documents at the very top of the ranked list.

5.7 Marginal Difference Analysis

Three system pairs exhibited MAP differences below 2%, requiring deeper analysis before making superiority claims:

System A	System B	Δ MAP	A wins	B wins	Ties
BM25+PRF	BM25	0.0083	92 queries	81 queries	52
BM25	LSA ($k=200$, CV)	0.0071	105 queries	83 queries	37
BM25+PRF	LSA ($k=200$, CV)	0.0154	110 queries	82 queries	33
VSM+QE	ESA (WordNet)	0.0184	91 queries	89 queries	45

Table 6: Query-level win/loss breakdown for system pairs with MAP difference < 2%. Note the last row: both QE and ESA-WordNet are below baseline, so “wins” here means beats the other degraded method, not beats the baseline.

The 0.71% MAP gap between BM25 and LSA translates to 105 vs. 83 per-query wins — a clear directional signal, but the gap is small enough that it does not survive as a decisive superiority claim without pair-level hypothesis testing. BM25+PRF wins on 110 queries against LSA (82), giving it the strongest per-query record among the top three methods. All three (BM25, BM25+PRF, LSA) represent genuinely comparable retrieval quality in the practical range.

Looking at the PRF-vs-BM25 pair (0.83% MAP gap) through the qrel density lens reveals *why* this marginal difference exists. PRF wins on 43 of 67 dense-qrel queries (64.2%) but only 5 of 35 sparse-qrel queries (14.3%). This follows directly from what PRF assumes: the first-pass BM25 top-3 are relevant enough to yield useful expansion terms. For dense queries with ≥ 10 relevant documents in the collection, this assumption usually holds — BM25 reliably surfaces at least one relevant document, and the expansion terms genuinely reflect the query topic. For sparse queries (1–3 relevant documents), first-pass reliability is lower, expansion terms come from irrelevant documents, and the second pass is actively misled — which is why plain BM25 wins 45.7% of sparse-query matchups vs. PRF’s 14.3%. The 0.83% aggregate MAP advantage for PRF is therefore a density-conditional effect: PRF is concretely better for dense queries, but slightly worse for sparse ones, and the Cranfield distribution (67 dense, 35 sparse) tips the average in PRF’s favour.

The BM25-vs-LSA gap (0.71% MAP) shows a different pattern. BM25 wins 53.7% of dense-query matchups vs. LSA’s 40.3%, and 40.0% vs. 25.7% for sparse. Unlike

dense-qrel
sparse-qrel

the PRF-vs-BM25 relationship, BM25’s advantage over LSA is present across *all* density classes and is actually larger for dense queries. This means LSA’s latent-space dimensionality reduction, while helpful for vocabulary matching, does not fully compensate for BM25’s superior term weighting (Robertson IDF, TF saturation) even when many relevant documents are available.

5.8 Per-Query Granularity Analysis

Aggregate MAP numbers conceal substantial variability across queries. Table 7 shows the number of queries on which each method improved, degraded, or produced no change relative to the baseline.

System	Queries Improved	Queries Degraded	Unchanged
BM25	142	37	46
BM25 + PRF	136	52	37
LSA ($k=200$, CV)	124	62	39
ESA (Cranfield)	120	71	34
ESA (WordNet)	60	120	45
VSM + QE	28	130	67

Table 7: Per-query win/loss counts against the VSM baseline ($k=10$). Rows sorted by queries improved.

BM25’s per-query profile is the most favourable: it improves 142 queries and degrades only 37, a ratio of nearly 4:1 in favour of improvement. BM25+PRF is second at 136/52. In contrast, VSM+QE degrades 130 queries while improving only 28 — a 1:4.6 ratio against improvement. This is a strong signal that unconstrained WordNet expansion introduces more noise than signal on this highly technical corpus, a finding now confirmed at the level of statistical significance by the two-sided Wilcoxon test.

5.9 Query-Class-Level Analysis

Aggregate MAP numbers treat all 225 queries as a flat pool. Here we break them down by two axes to see *for which class of query* each method wins or loses.

Classification scheme. Queries are classified by (1) length: short (≤ 5 words), medium (6–10 words), long (≥ 11 words); and (2) qrel density: sparse (1–3 relevant documents), medium (4–9), dense (≥ 10). Among 225 Cranfield queries, none has ≤ 5 words — all fall into medium (32 queries) or long (193 queries). By density: 35 sparse, 123 medium, 67 dense.

System	Medium (6–10 words)	Long (≥ 11 words)
VSM (Baseline)	0.2110	0.2787
BM25	0.2751	0.3396
BM25 + PRF	0.2831	0.3480
LSA ($k=200$, CV)	0.2735	0.3316
ESA (Cranfield)	0.2505	0.3040
VSM + QE	0.1964	0.2355
ESA (WordNet)	0.1506	0.2216

Table 8: Mean AP@10 by query length class. Sorted by long-query MAP. Bold entries are best per column.

The rank ordering is identical for both length classes (BM25+PRF > BM25 > LSA > ESA-Cranfield > Baseline > QE > ESA-WordNet). PRF’s lead over plain BM25 is somewhat more pronounced for medium-length queries (2.9% gap) than for long queries (2.5%), which is consistent with the idea that shorter queries give BM25’s first pass fewer terms to work with, so the PRF expansion term injection contributes more.

System	Sparse (1–3 rel.)	Medium (4–9 rel.)	Dense (≥ 10 rel.)
VSM (Baseline)	0.4195	0.2813	0.1680
BM25	0.4820	0.3383	0.2369
BM25 + PRF	0.4251	0.3557	0.2626
LSA ($k=200$, CV)	0.4672	0.3275	0.2405
ESA (Cranfield)	0.4273	0.3044	0.2132
VSM + QE	0.3589	0.2488	0.1280
ESA (WordNet)	0.3224	0.2199	0.1380

Table 9: Mean AP@10 by qrel density class. Bold entries are best per column. All systems score higher on sparse queries because reaching 1–3 relevant documents in top-10 is structurally easier than achieving recall over a 10+ relevant set.

The density breakdown exposes a leadership inversion: **BM25** leads the sparse-density class (0.4820) but **BM25+PRF** leads the medium and dense classes (0.3557 and 0.2626). This matches what we saw in the Marginal Difference Analysis: PRF’s expansion step depends on reliable first-pass retrieval, and that becomes more likely when many relevant documents are spread across the collection. The MAP numbers also drop consistently as density increases — this is a structural property of the metric itself. A query with only 2 relevant documents

needs just 2 hits in the top-10 for $\text{MAP} = 1.0$, while a query with 20 relevant documents can at best score around 0.49 at $k=10$ under perfect retrieval. So MAP comparisons across density classes need to account for this built-in ceiling difference.

5.10 Universal Failure Analysis

A **universal failure** is a query for which every evaluated system scores $\text{AP}@10 = 0$ — cases no implemented method can handle. We found 8 such queries out of 225 (3.6% of the collection).

QID	Query text (truncated)	Length	Density
9	papers on internal /slip flow/ heat transfer studies	medium	medium
22	did anyone else discover that the turbulent skin friction...	long	sparse
28	what application has the linear theory design of curved...	long	sparse
44	what are the details of the rigorous kinetic theory...	long	medium
62	how far around a cylinder and under what conditions...	long	medium
109	panels subjected to aerodynamic heating	medium	medium
207	how do large changes in new mass ratio affect wing-flutter	long	medium
216	what investigations have been made of the wave system...	long	sparse

Table 10: Universal failure queries: $\text{AP}@10 = 0$ for all seven systems. Classes from the query-class analysis.

Six of eight are long queries, and three are sparse-density (few relevant documents to find). The noteworthy outlier is Query 109 — “*panels subjected to aerodynamic heating*” — which is short (medium length class) and medium density. This query fails despite a reasonable structure because “panels” in the Cranfield corpus primarily refers to structural panels (thin plates under load) rather than the more common aeronautics sense; without any term that matches the relevant document vocabulary, no method in our set can recover. This shows that vocabulary mismatch — the same problem that failed the baseline — is still the root cause even after six different improvements. None of the methods in our set can recover when the query and the relevant documents simply do not share surface vocabulary and the corpus does not have enough co-occurrence signal to bridge the gap.

5.11 Systematic Failure Analysis

System	New Failures (vs. Baseline)	Baseline Failures Fixed
BM25	1	7
BM25 + PRF	4	5
LSA ($k=200$, CV)	5	6
ESA (Cranfield)	8	6
ESA (WordNet)	16	2
VSM + QE	16	0

Table 11: New failure cases introduced vs. baseline failures fixed, where a failure is defined as $AP@10 = 0$. The improved diagnostic in the code also records query length and relevant-doc count for each new failure.

A “new failure” means a query for which the baseline found at least one relevant document in the top-10 but the new method found none. VSM+QE introduces 16 new failures while fixing zero, confirming that query drift is a systematic problem. BM25 is the most conservative: only 1 new failure against 7 fixed — it changes the most failures in the correct direction while almost never breaking a query that was working.

Boundary Conditions: When Do Methods Introduce New Failures?

The table below breaks down new failures by query length and qrel density to identify the specific conditions under which each method degrades.

System	By query length		By qrel density		
	Medium	Long	Sparse	Medium	Dense
BM25	0	1	0	1	0
BM25 + PRF	1	3	1	3	0
LSA ($k=200$)	0	5	0	3	2
ESA (Cranfield)	0	8	3	4	1
ESA (WordNet)	1	15	3	7	6
VSM + QE	2	14	2	6	8

Table 12: New failures broken down by query length and qrel density class. No method introduces failures on short queries because no Cranfield query has ≤ 5 words after preprocessing.

Long queries (≥ 11 words) account for the vast majority of new failures across all methods. BM25’s single new failure is a 13-word acoustic-propagation query with medium density — a topic whose vocabulary is far enough from mainstream aerodynamics that BM25’s term weighting still falls short. VSM+QE introduces 8 new failures on dense-qrel queries and 6 on medium, confirming that vocabulary drift hurts most when many relevant documents exist and small ranking errors cascade into large recall losses.

5.12 Qualitative Case Study

We focus on the documented vocabulary mismatch failure: Query 1 and Document 29.

Query: “*what similarity laws must be obeyed when constructing aeroelastic models of heated high speed aircraft*”

Document 29 (relevant, qrel position 2) discusses “*transient temperatures and thermal stresses in aircraft structures subjected to aerodynamic heating*”.

System	Rank of Doc 29
VSM (Baseline)	109
BM25	79
LSA ($k=200$, CV)	77
VSM + QE	117 (worse than baseline)
BM25 + PRF	39
ESA (WordNet)	126 (worst)
ESA (Cranfield)	27 (best)

Table 13: Rank assigned to Document 29 for Query 1 by each system.

Tracing the rank of Document 29 across all seven systems shows exactly why each method either helps or fails here:

BM25 improves over VSM (65 vs. 109) because its Robertson IDF gives moderate-frequency terms like *aircraft* more discriminative weight, and its length normalisation prevents short abstracts from being systematically outscored by longer ones. However, it still cannot bridge the *heated/thermal* lexical gap.

LSA (rank 77) provides a partial but incomplete correction. The $k=200$ latent space captures some of the co-occurrence signal between heat-related terms, but the aerodynamics corpus is narrow enough that even 200 components do not fully separate the thermal cluster.

BM25+PRF (rank 34) achieves the second-best result. The first-pass BM25 retrieves documents that use *thermal* and *aerodynamic*, and the PRF expansion

injects these terms into the second-pass query, effectively bridging the lexical gap through an unsupervised expansion step.

ESA-Cranfield (rank 27) is the best performer for this specific query. By projecting both the query and documents into the full Cranfield document-level concept space, it captures the fact that Document 29 has a high similarity profile with documents that are themselves relevant to heated aerodynamic structures.

VSM+QE (rank 117, worse than baseline) exemplifies query drift. The primary WordNet synset for *heated* or *aeroelastic* likely retrieves semantically shallow synonyms that actively mislead the TF-IDF ranking, pushing Document 29 even further down.

5.13 Discussion: Why the Methods Work (or Fail)

Looking at all the results together, the main pattern that comes through is whether a method's underlying assumptions actually fit the Cranfield domain or not.

BM25 improved over VSM mostly because it adds TF saturation (a word appearing 50 times is not treated as $50\times$ more important than one appearing once) and length normalisation (short dense abstracts are not unfairly penalised against longer documents). The CV-selected $k_1=2.0$ and $b=0.9$ are higher than the standard defaults of 1.5/0.75, suggesting that on the Cranfield corpus — which has high variance in abstract length — stronger length normalisation is needed. Neither of these improvements requires any semantic knowledge of the domain, so they work cleanly.

LSA improved because the Cranfield corpus is all from one narrow domain, aerodynamics and fluid mechanics. This means the word co-occurrence patterns in the 1400 documents are fairly consistent, and the SVD at 200 components can pull out useful latent structure. The ablation showed that going beyond 200 components starts hurting because too many noisy dimensions get included.

BM25 with PRF worked reasonably well when the first BM25 pass returned documents that were in the same topic cluster as the relevant ones. The problem is when the first pass goes wrong — then the expansion terms come from irrelevant documents and the second pass gets actively misled. This happened enough times to prevent it from clearly beating LSA.

WordNet Query Expansion hurt performance. The reason is that Cranfield uses highly specialised aerodynamics vocabulary. Terms like *aeroelastic*, *blasius equation*, or *joule heating* either do not appear in WordNet at all, or they map to general-English synsets that are not useful here. For instance, expanding *viscous* with everyday synonyms for thick or sticky adds noise that pushes relevant documents down in the ranking.

ESA with WordNet concepts failed for essentially the same reason as QE: WordNet was built for everyday English, not for a 1950s aerodynamics research

collection. Terms central to Cranfield like *laminar*, *stagnation point*, or *oblique shock* are either not in WordNet or are too weakly connected there to be useful concept dimensions.

ESA with Cranfield concepts worked better than WordNet ESA because the concept space is built directly from the same corpus. Every specialised aerodynamics term naturally gets a concept vector that reflects how it is used in context of these 1400 documents. It is not the best method overall because the concept space is also noisy and quite high-dimensional, but at least the domain mismatch problem is avoided.

5.14 Summary of Findings

1. All six methods are statistically significantly different from the VSM baseline (Wilcoxon, $p < 0.05$). Four improve it; two degrade it.
2. BM25 ($k_1=2.0$, $b=0.9$, CV-selected) and BM25+PRF are the most practically reliable: highest per-query improvement rates (142 and 136 improved queries), fewest new failures (1 and 4), and strong aggregate metrics.
3. The MAP gap between BM25 (0.3304), BM25+PRF (0.3388), and LSA (0.3233) is marginal ($< 1.6\%$) and does not constitute a credible superiority claim without pair-level tests. Per-query analysis shows BM25+PRF wins on more individual queries than BM25.
4. VSM+QE and ESA-WordNet are statistically significantly *worse* than the baseline, not merely indistinguishable — a distinction only visible when using the correct two-sided Wilcoxon test.
5. LSA n_components is best selected by 5-fold CV ($k=200$, CV MAP = 0.3233). Selecting k on all 225 test queries constitutes data leakage; CV gave the same answer here, but the methodology matters.
6. MRR and graded nDCG provide complementary insight: BM25 leads on first-relevant-position quality (MRR = 0.8002); BM25+PRF leads on surfacing the most highly relevant documents (graded nDCG = 0.4887).
7. The vocabulary mismatch case study confirms that all three corpus-based methods (BM25, LSA, PRF) partially address the core failure, but none fully resolves it.

6 Conclusion

This project built and evaluated an Information Retrieval system on the Cranfield aerodynamics dataset, starting from a TF-IDF Vector Space Model baseline and testing six alternative retrieval approaches. The baseline VSM achieved MAP@10 = 0.2691 and nDCG@10 = 0.4203. The main problem identified was the orthogonality assumption, which caused vocabulary mismatch failures like Document 29 being placed at rank 109 for Query 1 despite being clearly relevant.

Six alternative systems were tested. BM25 ($k_1=2.0$, $b=0.9$, hyperparameters selected by 5-fold cross-validation) gave the most consistent improvement: MAP@10

= 0.3304 (+22.8% relative), Wilcoxon $p = 3.3 \times 10^{-16}$, Cohen's $d = 0.57$ (medium effect), improving 142 out of 225 queries and degrading only 37. BM25+PRF achieved MAP@10 = 0.3388 with the best graded nDCG (0.4887), and LSA at $k=200$ (selected by 5-fold CV to avoid data leakage) achieved MAP@10 = 0.3233. All three are statistically significantly better than the baseline.

Importantly, the correct application of two-sided Wilcoxon tests for VSM+QE and ESA-WordNet revealed that they are not merely indistinguishable from the baseline — they are statistically significantly *worse* ($p < 10^{-8}$). This distinction matters: one-sided tests had previously masked the harm these methods cause.

The overall takeaway is that methods which rely on the corpus itself (BM25's statistics, LSA's SVD, PRF's feedback) worked, while methods that import outside knowledge from WordNet failed because the domain mismatch was too large. One direction to explore further would be transformer-based dense retrieval, which learns semantic representations directly from text rather than depending on a hand-built lexical resource like WordNet.